

Error-Aware Reviewer Guidance for AI-Assisted Fisheries Electronic Monitoring

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The problem

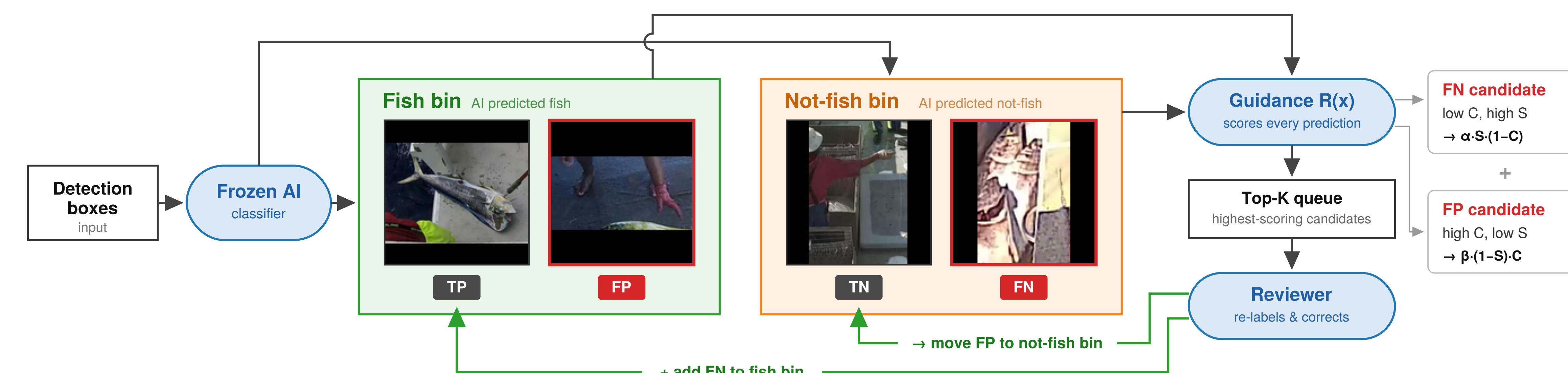
- Deck cameras record 100% of fishing activity. Reviewers watch about 10%.
- AI can watch all of it, but its predictions are wrong sometimes and nobody knows where.
- Errors outside the reviewed slice stay in the record.

Given a fixed 10% budget, which 10% should a human look at?

No retraining. No larger model. The only change is where the reviewer looks.

Evaluated on FDA-D, 200 sablefish clips with 57 test events, and Fishnet, 143,818 deck images.

Error-aware reviewer guidance



$$R(x) = \alpha S(x)(1 - C(x)) + \beta(1 - S(x))C(x)$$

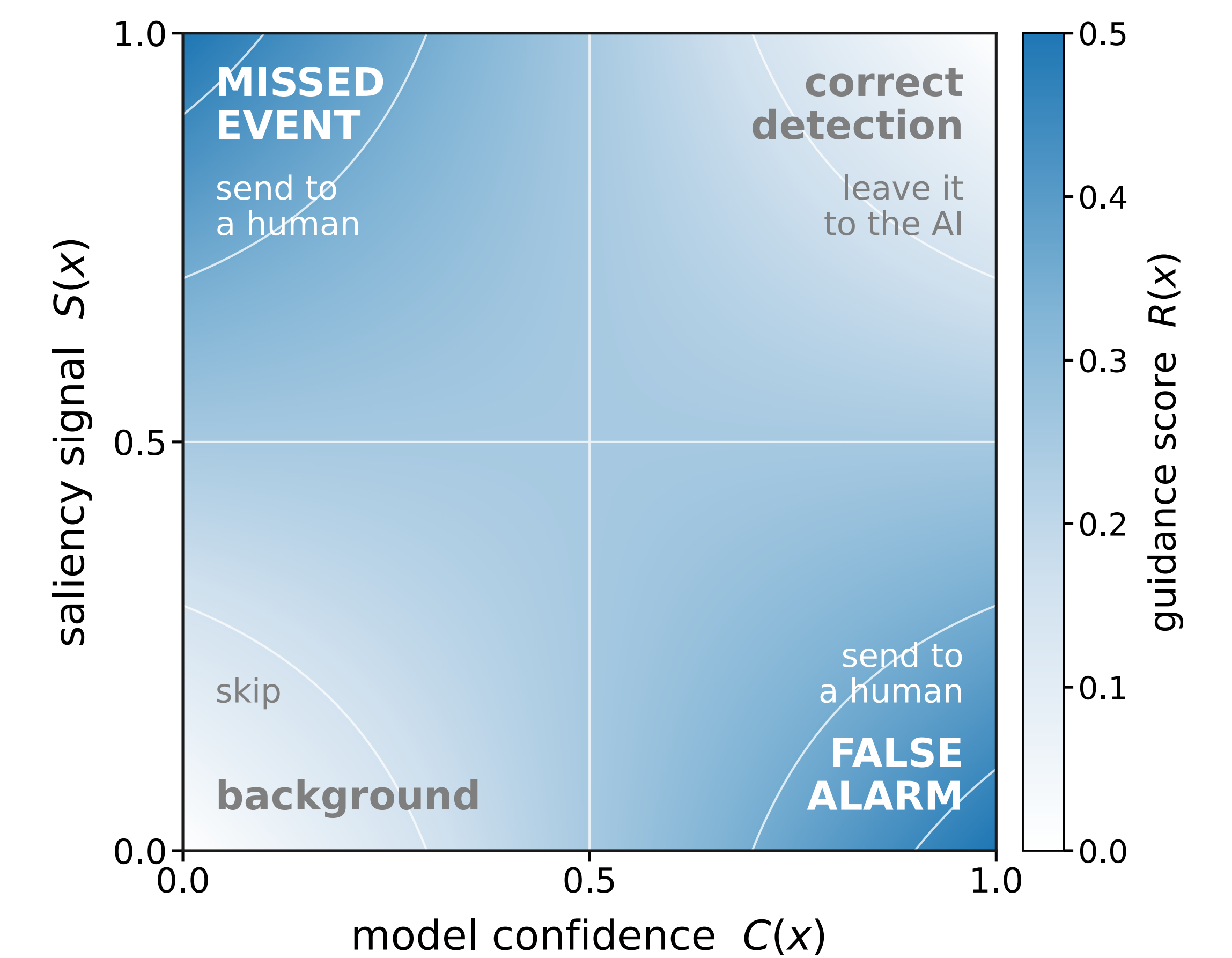
$\alpha + \beta = 1$, user-defined. We use the neutral default 0.5.

$S(x)$: evidence that the unit contains an event, computed without using the detector's confidence.

$C(x)$: the detector's own calibrated score for that unit. Both lie in $[0, 1]$.

Rank by $R(x)$, review the top K , apply the corrections. The model stays frozen: no retraining, no failure labels, no extra budget.

Where the score is high



The two terms are added, not multiplied, so the score peaks at both error corners rather than in the middle where neither error is indicated.

Same budget, better record

50 → 55 of 57

events recovered at the same 10% budget.

Guided review (ours)

5 of 7

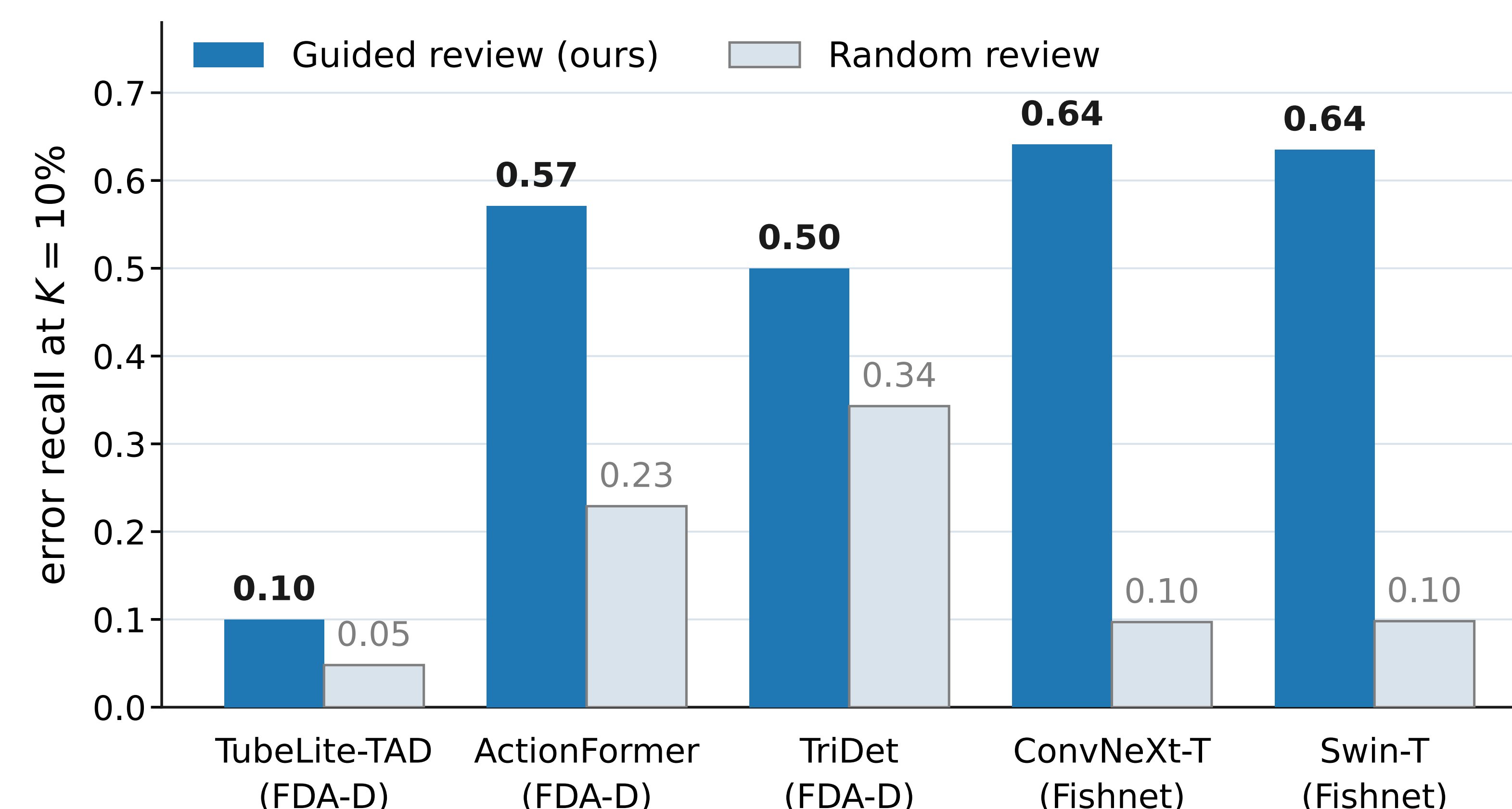
Random review

2 of 7

Each circle is one event the detector missed on the FDA-D test split. Filled circles reach the reviewer at a 10% budget.

Strategy ($K = 10\%$)	Events /57	Event recall	Box F1	Fish recall
No AI, 10% random review	5.7	0.100	0.182	0.100
AI alone	50	0.877	0.973	0.958
AI + 10% random review	52.0	0.912	0.976	0.962
AI + 10% guidance (ours)	55	0.965	0.990	0.984
Oracle (100% review)	57	1.000	1.000	1.000

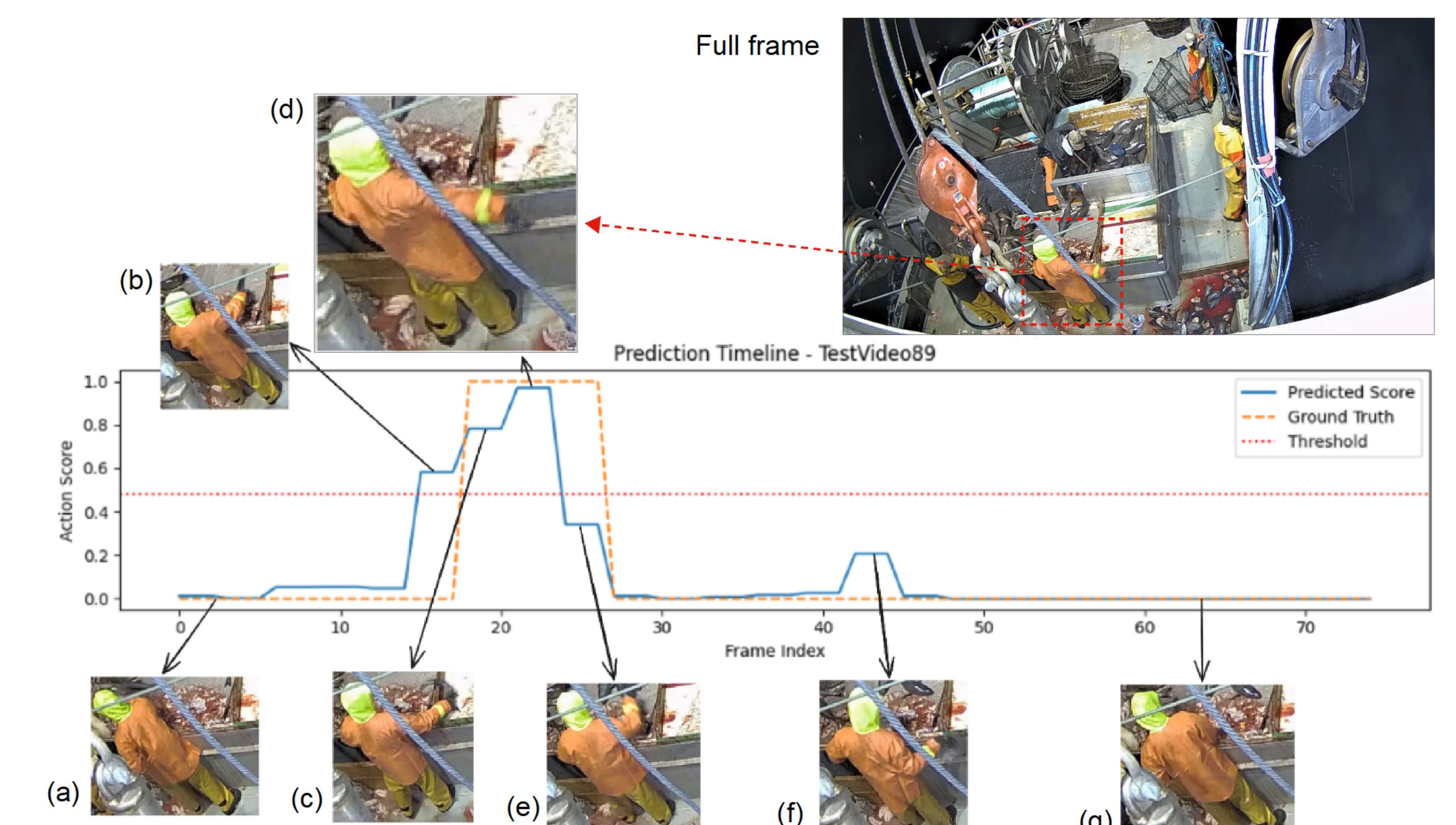
It transfers across models



Three temporal detectors and two image classifiers. Guidance reaches more errors than random review in every case, and all five improvements have 95% bootstrap intervals that exclude zero.

The framework transfers across output structures. The choice of saliency signal does not: peaked confidence admits a model-output signal, flat output needs internal or external evidence.

Why the signal helps



Averaging confidence over a window discards its local shape. A window at an action boundary and a flat background window can share the same mean, so the aggregate cannot separate them. Local contrast can.

Limitations. One commercial fishery and one deck-imagery dataset. Reviewers are assumed to correct every error they are shown.